Sitong Fang

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Foundation model training for physical intelligence.
Research interests: world models and trustworthy multimodal AI.

Education

Peking University B.S., 2023 – 2027 (expected)
Yuanpei College, Artificial Intelligence

- **Honors:** SenseTime Scholarship; Yuanpei Young Scholar; Boya Scholarship; Soong Ching Ling Future Scholarship; Beijing Natural Science Foundation *QiYan* Research Program; Freshman Scholarship (First Prize)
- **Technical Skills:** Python, PyTorch, vLLM, DeepSpeed, Ray, CUDA, C/C++, LaTeX

Publications

When Slower Isn't Truer: Inverse Scaling Law of Truthfulness in Multimodal Reasoning **ACL 2026**
Sitong Fang, Wenjing Cao, Jiahao Li, Xuyao Wang, Chi-Min Chan, Sirui Han, Juntao Dai, Yike Guo, Yaodong Yang, Jiaming Ji

- Discovered an inverse scaling law: slower reasoning models are less truthful in multimodal settings.
- Proposed **TruthfulVQA**, the first benchmark for multimodal truthfulness evaluation with 5,000+ images, and **TruthfulJudge**, a human-in-the-loop evaluation framework.

Debate with Images: Detecting Deceptive Behaviors in Multimodal Large Language Models **ICML 2026**
Sitong Fang, Shiyi Hou, Kaile Wang, Boyuan Chen, Donghai Hong, Jiayi Zhou, Juntao Dai, Yaodong Yang, Jiaming Ji

- Introduced **MM-DeceptionBench**, the first benchmark for evaluating deceptive behaviors in multimodal LLMs.
- Proposed **Debate with Images**, a multi-agent debate framework requiring models to ground claims in visual evidence, significantly improving deception detection accuracy.

AI Alignment Can Build Moral Autonomy **NeurIPS 2026**
Kaile Wang*, Hantao Lou*, Tianyi Qiu*, Sebastian Sunday-Grève, Sitong Fang, Xinxiong Wu, Jiaming Ji, Yaodong Yang * Equal contribution

- A position paper on moral autonomy as an alignment goal, grounded in reflective endorsement, articulated principles, and practical wisdom.

A Blind Spot in Alignment: Quantifying Biosecurity Risks in Large Language Models **COLM 2026**
Shu Quan, Tianfang Hao, Sitong Fang, He Geng, Jiayi Zhou, Boyuan Chen, Kaile Wang, Donghai Hong, Juntao Dai, Yaodong Yang, Jiaming Ji

- Evaluating biosecurity risks in language models with **SPIKE-Bench** and the Functional Harmfulness Rate.

AI Deception: Risks, Dynamics, and Controls
Boyuan Chen*, Sitong Fang*, Jiaming Ji*, …, Yaodong Yang†, Tiejun Huang†, Ya-Qin Zhang†, HongJiang Zhang†, Andrew Yao† * Equal contribution † Corresponding author

- The first systematic international report on AI deception, with Turing Award laureate Andrew Yao as a corresponding author.
- Formally defined AI deception using signaling theory, analyzed the deception cycle, and proposed mitigation strategies.

Mitigating Deceptive Alignment via Self-Monitoring **Preprint**
Jiaming Ji*, Wenqi Chen*, Kaile Wang, Donghai Hong, Sitong Fang, Boyuan Chen, Jiayi Zhou, Juntao Dai, Sirui Han, Yike Guo, Yaodong Yang * Equal contribution

- Proposed **CoT Monitor+**, embedding a self-monitor in chain-of-thought reasoning to detect and suppress deceptive alignment.

Research Experience

Physis AI Feb 2026 – Present
Partner

- Training foundation models that learn physical dynamics and predict the consequences of actions.

PKU-Alignment Group, Peking University Dec 2024 – Jan 2026
Researcher · Advised by Prof. Yaodong Yang

- Led the development of **TruthfulVQA**, the first benchmark for multimodal truthfulness evaluation (ACL 2026).
- Led the development of **Debate with Images** (ICML 2026), a visually grounded multi-agent debate framework for multimodal deception detection, along with **MM-DeceptionBench**, the first benchmark for multimodal deception.
- Co-first authored the **AI Deception Survey**, with Turing Award laureate Andrew Yao as corresponding author.
- Core contributor and maintainer of **Align-Anything**, a multimodal post-training framework, and **Eval-Anything**, a multimodal evaluation harness.

HKGAI R&D Center / HKUST Dec 2024 – Mar 2025
Researcher

- Contributed to training **HKGAI-V1**, a multilingual large language model tailored to Hong Kong, and received a letter of acknowledgement from the Hong Kong Generative AI R&D Center.

Honors and Awards

2026 SenseTime Scholarship *30 recipients nationwide*
2025 Yuanpei Young Scholar, Peking University *10 annual recipients*
2025 Beijing Natural Science Foundation Undergraduate *QiYan* Research Program
Sole recipient among PKU AI undergraduates admitted in 2023
2025 Soong Ching Ling Future Scholarship
2024 Boya Scholarship, Peking University
2024 CMSC Scholarship, Peking University
2024 Academic Excellence Award, Peking University
2024 Social Service Award, Peking University
2023 Freshman Scholarship (First Prize), Peking University
2023 1st in Fujian Province, National College Entrance Examination (Physics Track)